

Dealing with Outliers: Identifying and Managing Extreme Values

Outliers—data points that differ significantly from other observations—can distort analysis and lead to flawed trading decisions. They might represent:

- Actual market events (flash crashes, major news)
- Data errors
- Corporate actions (splits, dividends)
- Thin market conditions

Here's how to identify and handle outliers:

```
1. import yfinance as yf
2. import pandas as pd
3. import numpy as np
4. import matplotlib.pyplot as plt
5.
6. # Download data
7. data = yf.download("AAPL", start="2023-01-01", end="2023-07-01")
8.
9. # Calculate daily returns
10. data['Returns'] = data['Close'].pct_change()
11.
12. # Identify outliers using z-score (standard deviations from mean)
13. z_scores = (data['Returns'] - data['Returns'].mean()) / data['Returns'].std()
14. outliers = abs(z_scores) > 3 # More than 3 standard deviations
15.
16. print("Number of outlier days:", outliers.sum())
17. print("Outlier dates:", data[outliers].index.tolist())
18.
19. # Options for handling outliers:
20.
21. # 1. Capping (winsorizing) - replace with threshold values
22. threshold = 3 * data['Returns'].std()
23. data['Returns_capped'] = data['Returns'].clip(
24.     lower=data['Returns'].mean() - threshold,
```